

NOTETAKERS NEEDED

**VOLUNTEER
TO BE A
NOTETAKER**

We are looking
for a volunteer
notetaker for this
class. If you would
like to volunteer
please email
notetaker@uwindSOR.ca

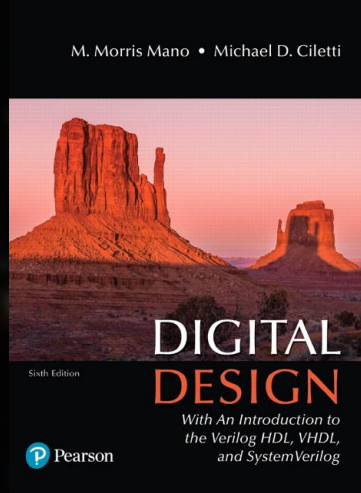




LEC02 & LAB02

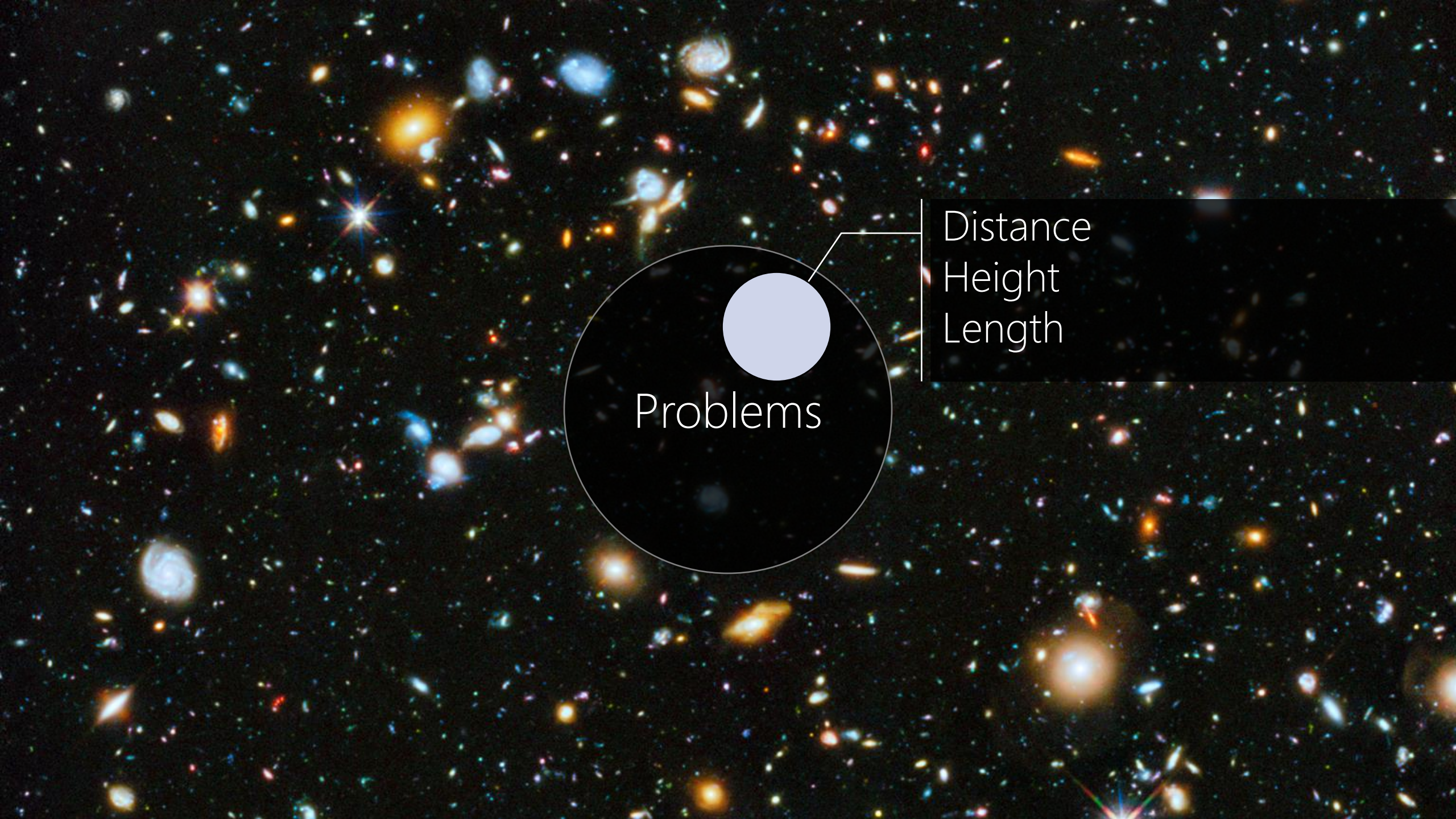
Week02 >> Lec02: Number Systems

Week02 >> Lab02: Programming Environment Setup



Chapter 1

Digital Systems and Binary Numbers

The background is a deep space photograph showing a dense field of galaxies in various colors (blue, orange, white) against a black sky. A diagram is overlaid on the right side. It consists of a large black circle with a smaller light blue circle inside it. A line connects the light blue circle to a black rectangular box on the right. Inside the large black circle, the word "Problems" is written in white. Inside the black rectangular box, the words "Distance", "Height", and "Length" are listed vertically in white.

Problems

Distance
Height
Length



Quantization



DISCRETE SYSTEMS

UNARY SYSTEM

aka. Base-1

I II III IV V VI VII VIII IX X

Roman Numerals
Originated in Ancient Rome
8th Century BC

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

NUMBER SYSTEMS

QUATERNARY SYSTEM

/kwaa·tur·neh·ree/

aka. Base-4, Radix-4

$(0,1,2,3)_4$

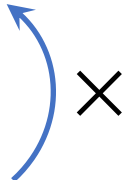
Hindu-Arabic Numerals
Originated in India
7th Century AD

More Significant Position
←

#Symbols=4
Radix-4
Base-4

4^3	4^2	4^1	4^0	
		2	0	$2 \times 4^1 + 0 \times 4^0 = 8$
		2	1	$2 \times 4^1 + 1 \times 4^0 = 9$
		2	2	$2 \times 4^1 + 2 \times 4^0 = 10$
		2	3	$2 \times 4^1 + 3 \times 4^0 = 11$
		3	0	$3 \times 4^1 + 0 \times 4^0 = 12$
		3	1	$3 \times 4^1 + 1 \times 4^0 = 13$
		3	2	$3 \times 4^1 + 2 \times 4^0 = 14$
		3	3	$3 \times 4^1 + 3 \times 4^0 = 15$
	1	0	0	$1 \times 4^2 + 0 \times 4^1 + 0 \times 4^0 = 16$



4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0	
3	0	3	0	2	1	3	1	
3×4^7	0×4^6	3×4^5	0×4^4	2×4^3	1×4^2	3×4^1	1×4^0	Σ
								65,437

What is the max number given n position?

$$N = 3 \times 4^{n-1} + 3 \times 4^{n-2} + \dots + 3 \times 4^2 + 3 \times 4^1 + 3 \times 4^0$$

$$N = 3 \times (4^{n-1} + 4^{n-2} + \dots + 4^2 + 4^1 + 4^0) \rightarrow \text{geometric series}$$

$$N = 3 \times \left(\frac{4^n - 1}{4 - 1} \right)$$

$$N = 4^n - 1$$

$3n$
Hossein's
System

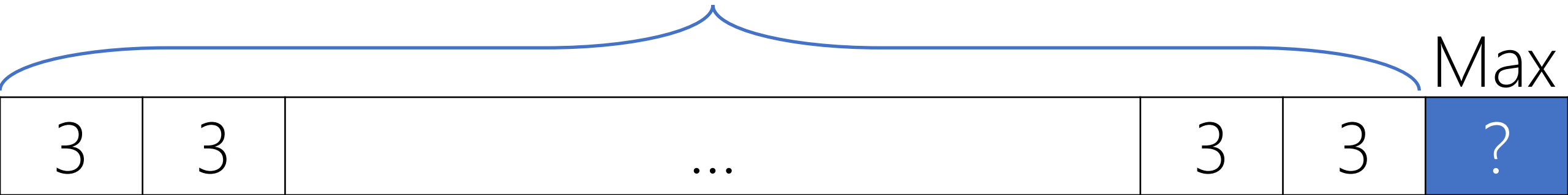
n positions

4^{n-1}	4^{n-2}					4^2	4^1	4^0	
3	3	...				3	3	3	N

What is the max number given n position?

$$n = 10 \Rightarrow 4^{10} - 1 = 1,048,575$$

10 positions



The other way around,
What is number of required position for a number N?

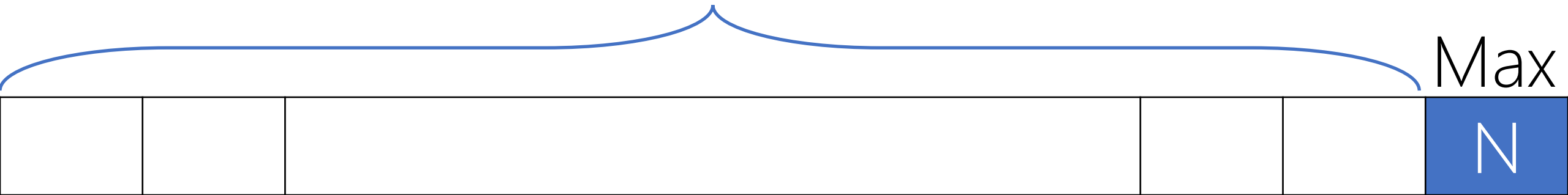
$$4^n - 1 = N$$

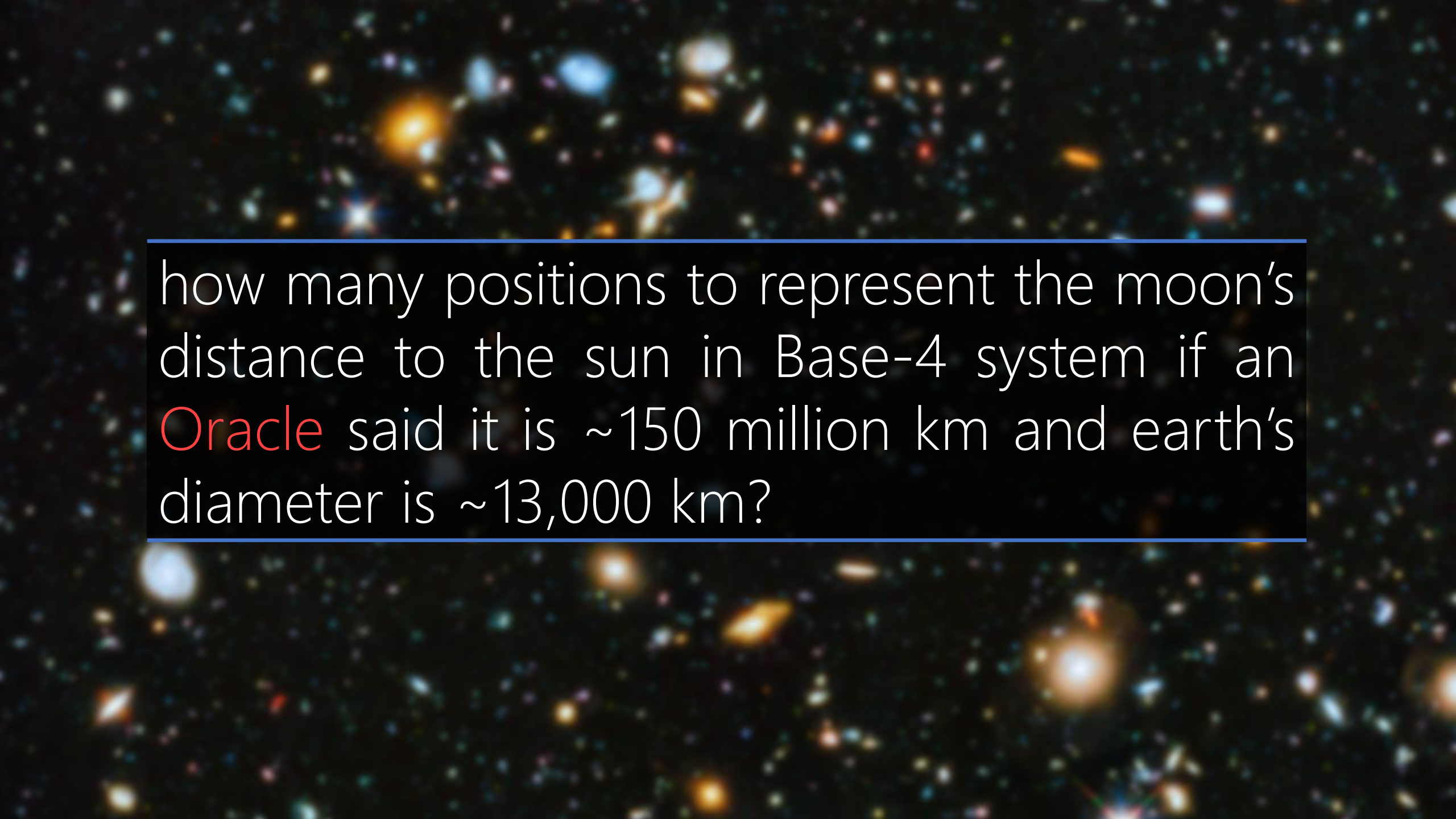
$$4^n = N + 1$$

$$\log_4 4^n = \log(N + 1)$$

$$n = \log_4(N + 1)$$

? positions





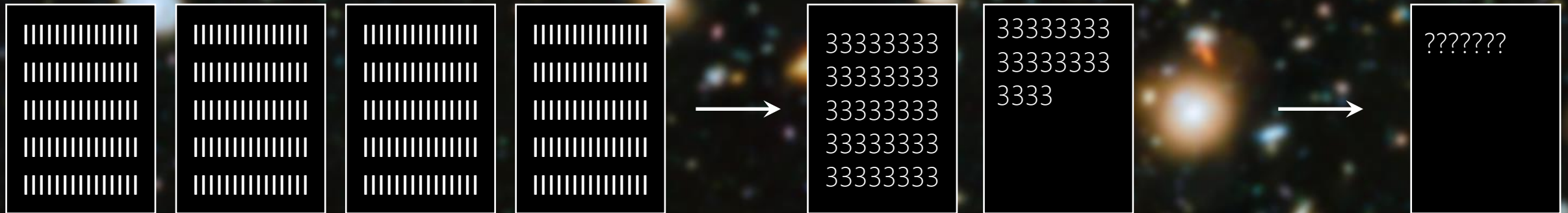
how many positions to represent the moon's distance to the sun in Base-4 system if an **Oracle** said it is ~150 million km and earth's diameter is ~13,000 km?

$\sim 150 \text{ million km} \div \sim 13,000 \text{ km} = \sim 12,000 \text{ Earth}$

$$N = 12,000$$

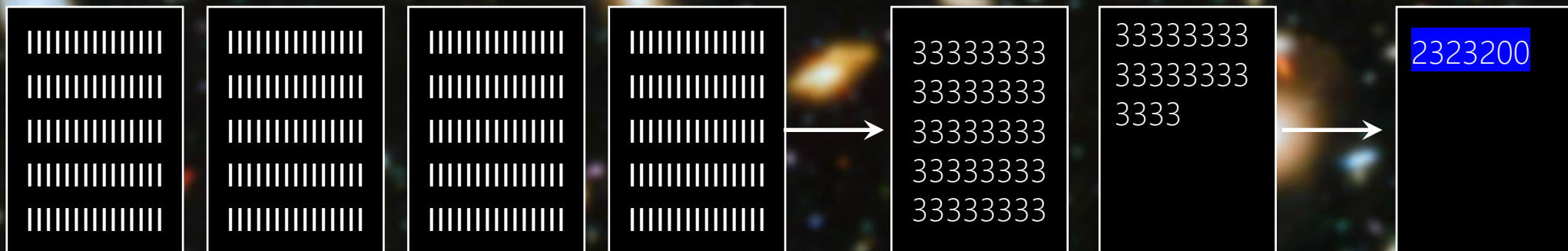
$$n = \text{Log}_4 (12,000+1) = \text{Log}_{10} 12,001 \div \text{Log}_{10} 4 = 4 \div 0.6 = 6.79$$

$\sim 7 \text{ positions}$



$$N = 12,000 \rightarrow (2323200)_4$$

We'll see how to convert from decimal to base-4 or any other number systems later. Stay tuned!

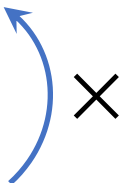


A deep-field astronomical image showing a vast field of galaxies in various colors (yellow, orange, blue, red) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, diffuse clouds and others as more compact, point-like sources. Two thin, horizontal blue lines are positioned above and below the central text.

COMMON NUMBER SYSTEMS

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines are positioned above and below the central text.

BINARY | BASE-2 | RADIX-2
 $(0,1)_2$

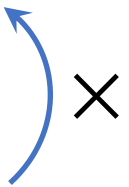
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	Σ
								173

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
3	0	3	0	2	1	3	1	
								Σ

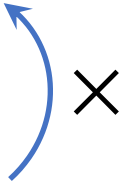
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
3	0	3	0	2	1	3	1	
-	0×2^6	-	0×2^4	-	1×2^2	-	1×2^0	Σ
								Not Valid

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, diffuse clouds and others as more compact, point-like sources.

OCTAL | BASE-8 | RADIX-8
(0,1,2,3,4,5,6,7)₈

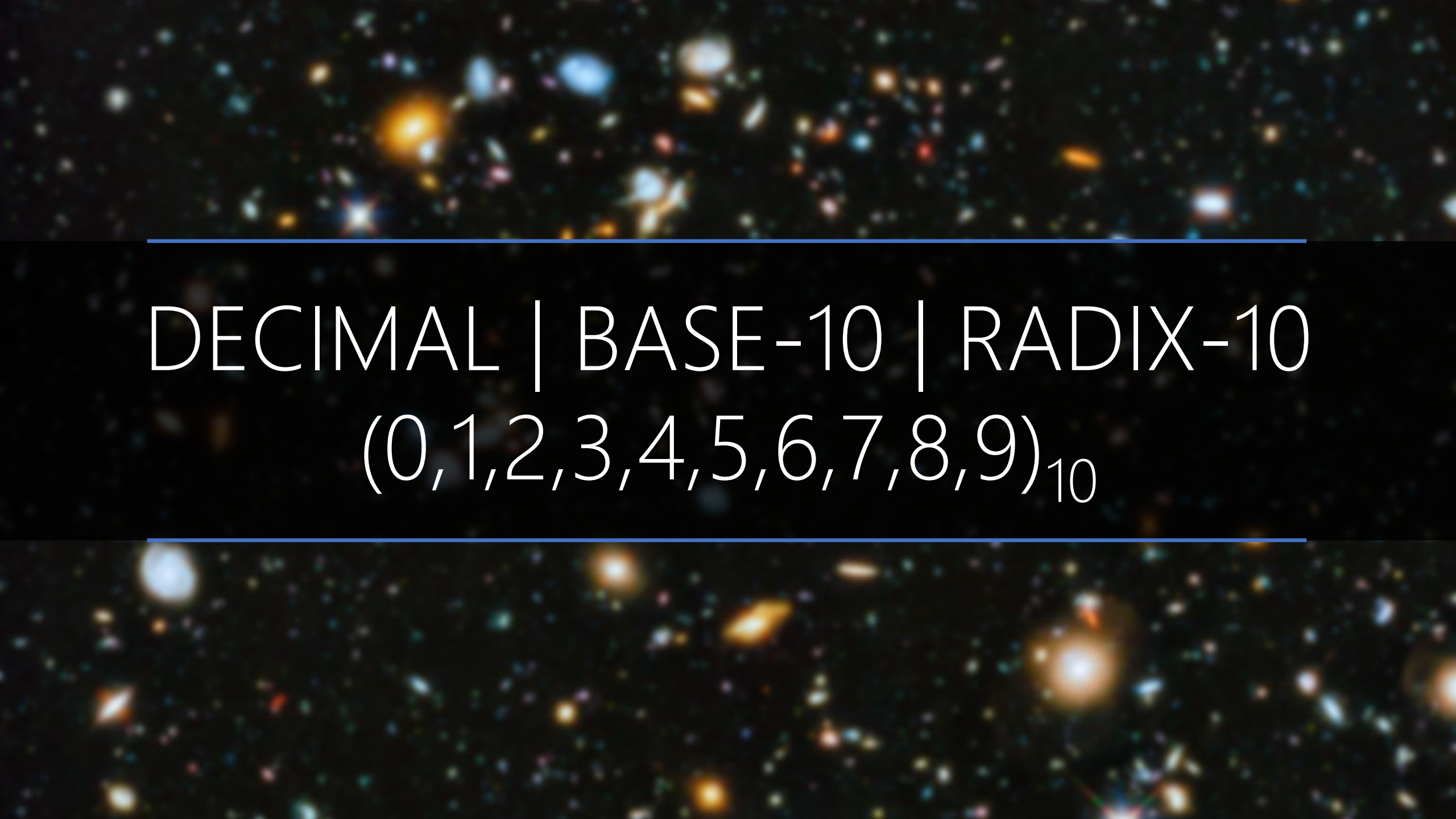
8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	3	0	2	1	3	1	
3×8^7	0×8^6	3×8^5	0×8^4	2×8^3	1×8^2	3×8^1	1×8^0	Σ
								6,390,873

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	A	0	8	1	3	1	Σ

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	A	0	8	1	3	1	
3×8^7	0×8^6	-	0×8^4	-	1×8^2	3×8^1	1×8^0	Σ
								Not Valid

Radix-8
vs.
Radix-2

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	
								173

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

DECIMAL | BASE-10 | RADIX-10
(0,1,2,3,4,5,6,7,8,9)₁₀

Arabic

Eastern Arabic

0123456789

٠ ١ ٢ ٣ ٤ ٥ ٦ ٧ ٨ ٩

*"the **hand** makes the two complementary aspects of integers entirely intuitive. It serves as an instrument permitting natural movement between cardinal and ordinal numbering. If you need to show that a set contains three, four, seven or ten elements, you raise or bend **simultaneously** three, four, seven or ten fingers, using your hand as cardinal mapping. If you want to count out the same things, then you bend or raise three, four, seven or ten fingers in **succession**, using the hand as an ordinal counting tool."*

- Georges Ifrah The Universal History of Numbers (Wiley, 2000, pp. 21-22)

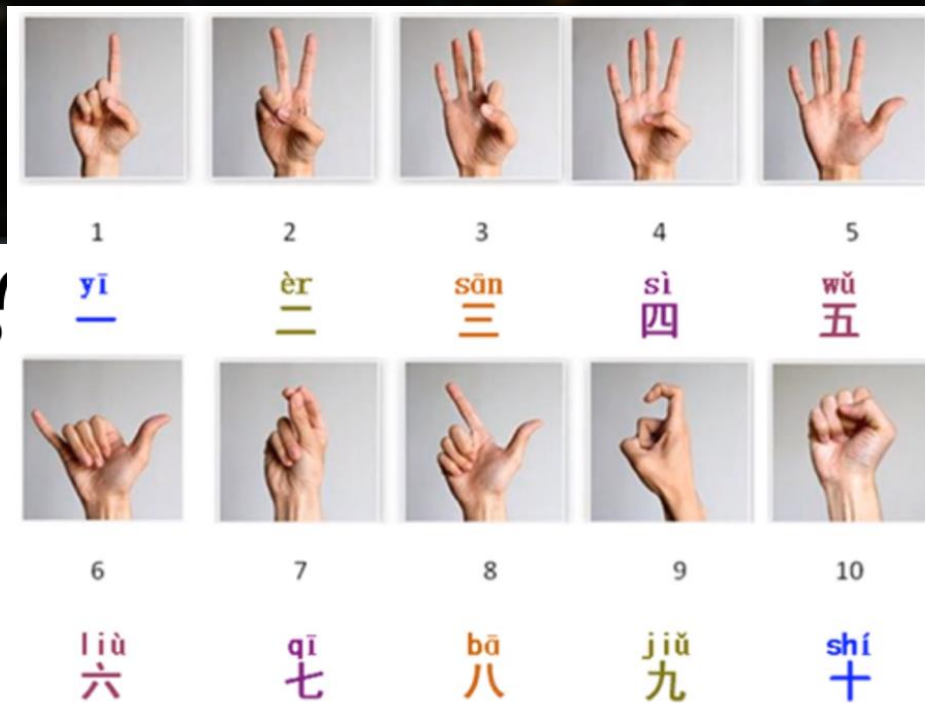
Thanks Cecelia Nydam and Giavi Tran for the hint!

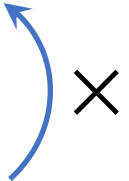
Thai

Chinese

๐ ๑ ๒ ๓ ๔

〇 一 二 三

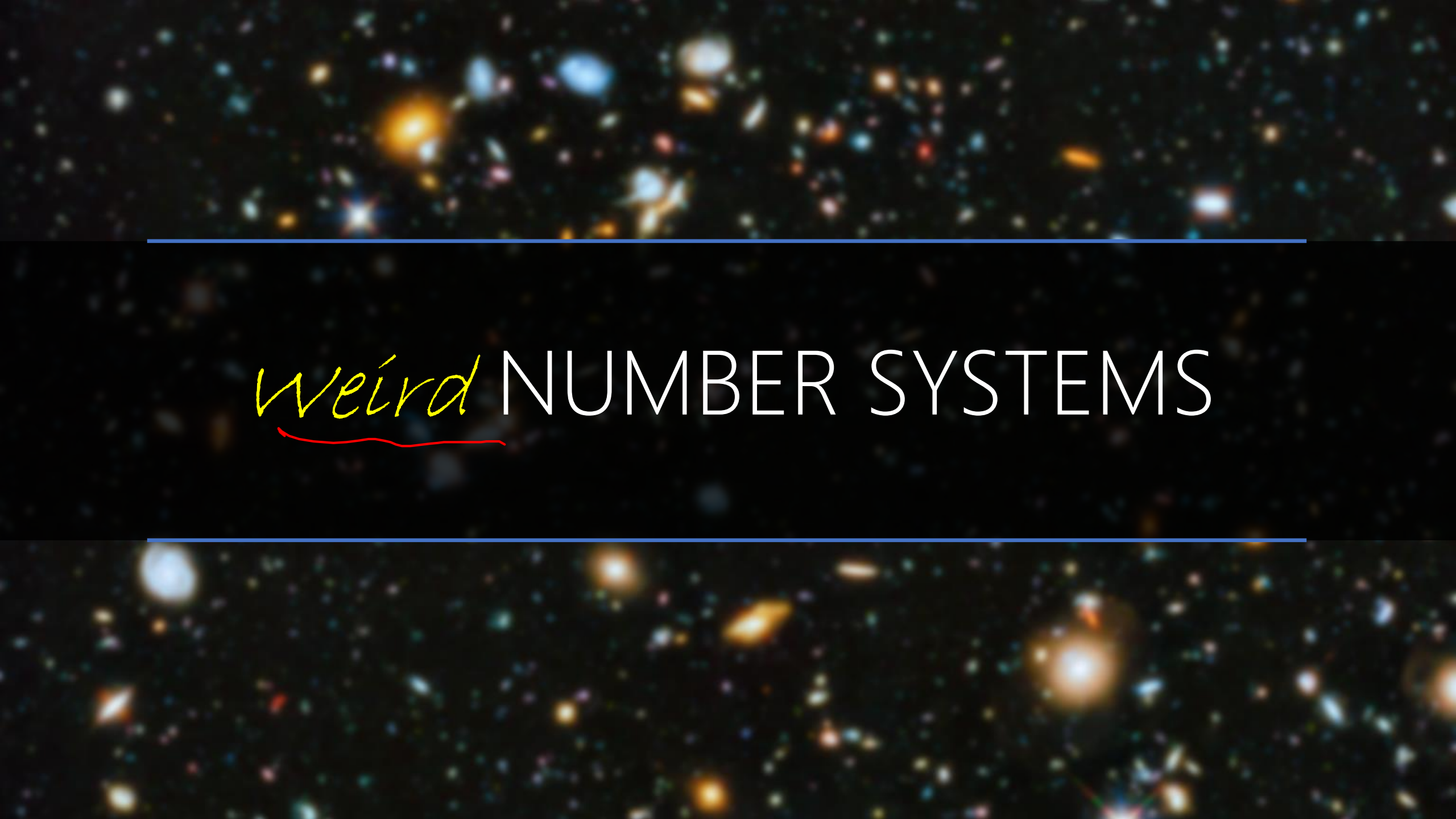


10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
1	0	1	0	1	1	0	1	
1×10^7	0×10^6	1×10^5	0×10^4	1×10^3	1×10^2	0×10^1	1×10^0	Σ
								10,101,101

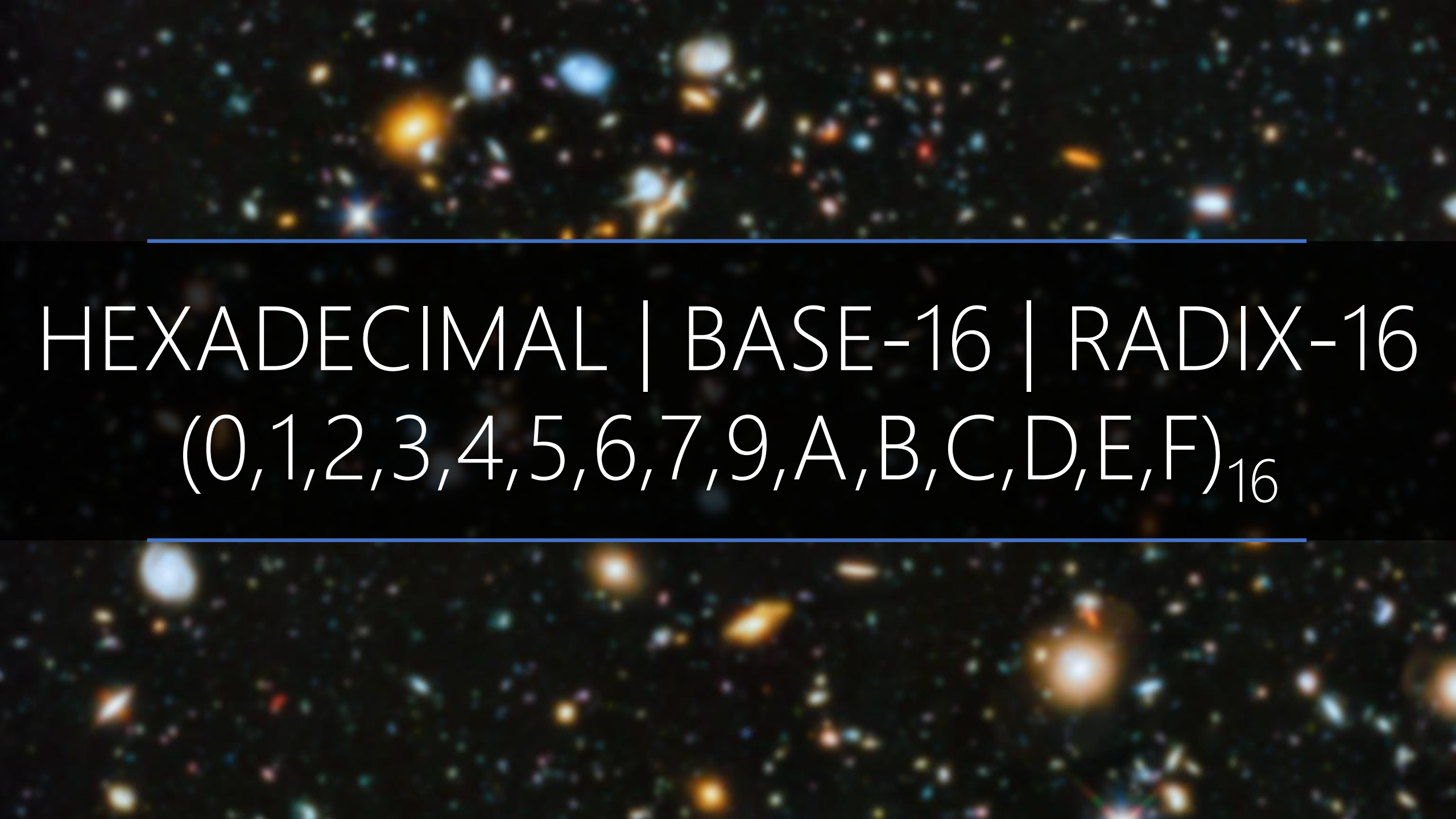
10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
1	0	1	0	1	1	0	1	
1×10^7	0×10^6	1×10^5	0×10^4	1×10^3	1×10^2	0×10^1	1×10^0	
								10,101,101

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
1	0	1	0	1	1	0	1	
1×8^7	0×8^6	1×8^5	0×8^4	1×8^3	1×8^2	0×8^1	1×8^0	
								2,130,497

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	
								173

A deep-field astronomical image showing a vast field of galaxies in various colors (yellow, orange, blue, red) against a black background. Two horizontal blue lines frame the central text.

Weird NUMBER SYSTEMS

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a dark space. A solid black horizontal band runs across the middle of the image, serving as a background for the text.

HEXADECIMAL | BASE-16 | RADIX-16
(0,1,2,3,4,5,6,7,9,A,B,C,D,E,F)₁₆

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	 $\times$
3	0	A	0	9	1	3	1	
3×16^7	0×16^6	$A \times 16^5$	0×16^4	9×16^3	1×16^2	3×16^1	1×16^0	Σ
								815,829,297

 $A = (9 + 1) = (10)_{10}$

$$1, 2, 3, 4, 5, 6, 7, 8, 9, A = 9 + 1 = (10)_{10}$$

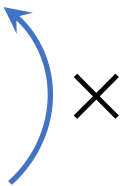
$$B = A + 1 = (11)_{10}$$

$$C = B + 1 = (12)_{10}$$

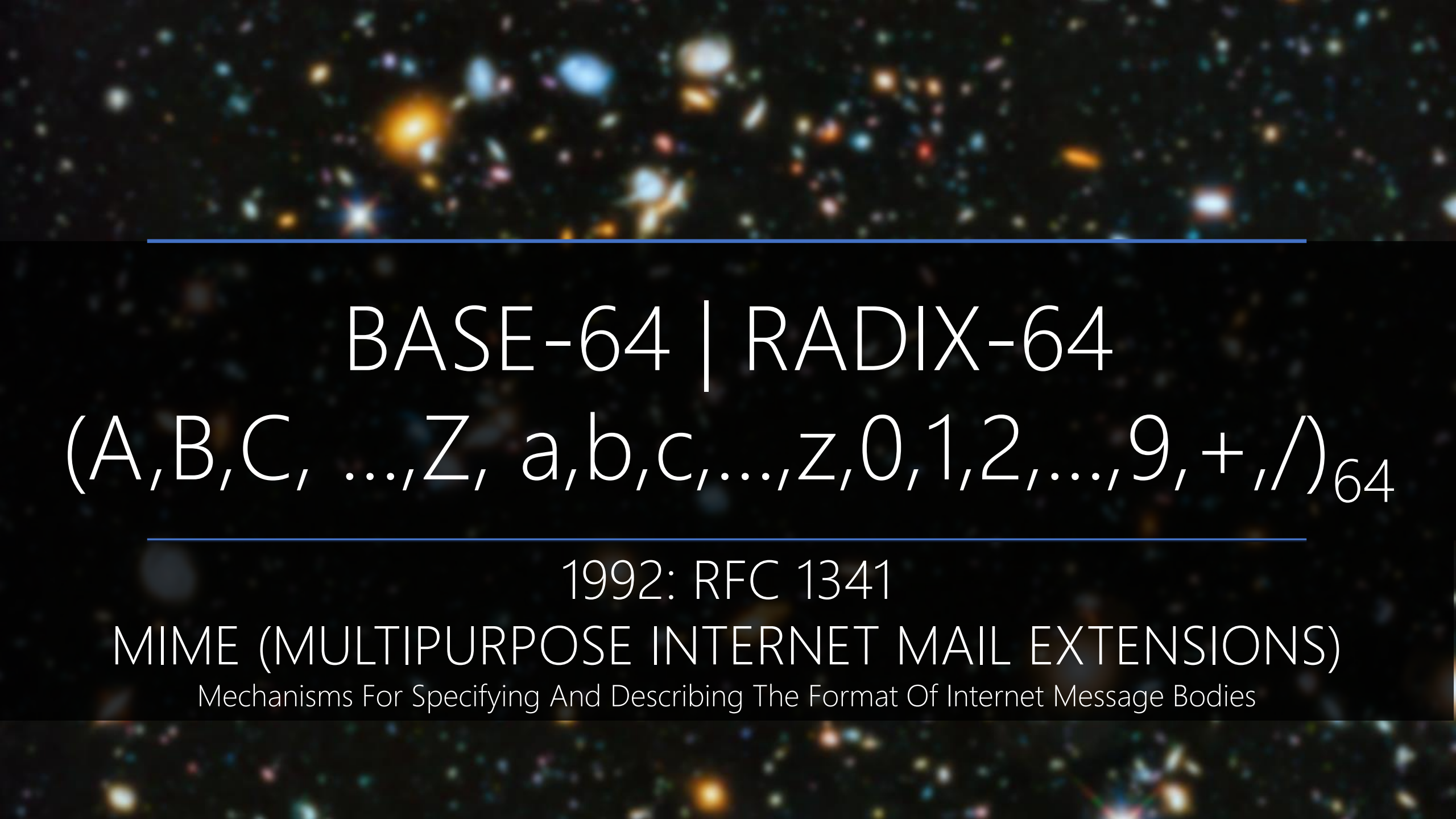
$$D = C + 1 = (13)_{10}$$

$$E = D + 1 = (14)_{10}$$

$$F = E + 1 = (15)_{10}$$

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
1	0	1	0	1	1	0	1	
1×16^7	0×16^6	1×16^5	0×16^4	1×16^3	1×16^2	0×16^1	1×16^0	Σ
								269,488,385

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
1	0	1	0	1	1	0	1	
1×16^7	0×16^6	1×16^5	0×16^4	1×16^3	1×16^2	0×16^1	1×16^0	
								269,488,385
10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
1	0	1	0	1	1	0	1	
1×10^7	0×10^6	1×10^5	0×10^4	1×10^3	1×10^2	0×10^1	1×10^0	
								10,101,101
8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
1	0	1	0	1	1	0	1	
1×8^7	0×8^6	1×8^5	0×8^4	1×8^3	1×8^2	0×8^1	1×8^0	
								2,130,497
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	
								173



BASE-64 | RADIX-64
(A,B,C, ...,Z, a,b,c,...,z,0,1,2,...,9,+ ,/)₆₄

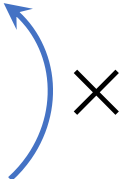
1992: RFC 1341

MIME (MULTIPURPOSE INTERNET MAIL EXTENSIONS)

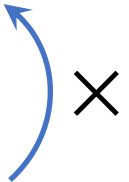
Mechanisms For Specifying And Describing The Format Of Internet Message Bodies

Digit	Value		Digit	Value		Digit	Value		Digit	Value
A	0		Q	16		g	32		w	48
B	1		R	17		h	33		x	49
C	2		S	18		i	34		y	50
D	3		T	19		j	35		z	51
E	4		U	20		k	36		0	52
F	5		V	21		l	37		1	53
G	6		W	22		m	38		2	54
H	7	→	X	23	→	n	39	→	3	55
I	8		Y	24		o	40		4	56
J	9		Z	25		p	41		5	57
K	10		a	26		q	42		6	58
L	11		b	27		r	43		7	59
M	12		c	28		s	44		8	60
N	13		d	29		t	45		9	61
O	14		e	30		u	46		+	62
P	15		f	31		v	47		/	63

3	a	A	/	d	1	H	+	
---	---	---	---	---	---	---	---	--

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
								Σ

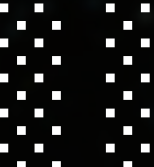
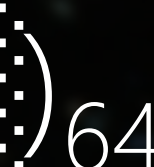
Digit	Value		Digit	Value		Digit	Value		Digit	Value
A	0	→	Q	16	→	g	32	→	w	48
B	1		R	17		h	33		x	49
C	2		S	18		i	34		y	50
D	3		T	19		j	35		z	51
E	4		U	20		k	36		0	52
F	5		V	21		l	37		1	53
G	6		W	22		m	38		2	54
H	7		X	23		n	39		3	55
I	8		Y	24		o	40		4	56
J	9		Z	25		p	41		5	57
K	10		a	26		q	42		6	58
L	11		b	27		r	43		7	59
M	12		c	28		s	44		8	60
N	13		d	29		t	45		9	61
O	14		e	30		u	46		+	62
P	15		f	31		v	47		/	63

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
55 $\times 64^7$	26 $\times 64^6$	0 $\times 64^5$	63 $\times 64^4$	29 $\times 64^3$	53 $\times 64^2$	7 $\times 64^1$	62 $\times 64^0$	Σ

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
55 $\times 64^7$	26 $\times 64^6$	0 $\times 64^5$	63 $\times 64^4$	29 $\times 64^3$	53 $\times 64^2$	7 $\times 64^1$	62 $\times 64^0$	Σ
								243,680,329, 290,238

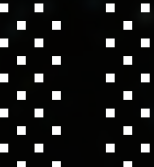
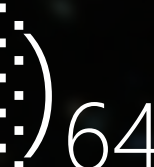


LET'S COUNT IN BASE-64

(   )₆₄



LET'S COUNT IN BASE-64

(   )₆₄

AAAA, AAAB, AAAC, ..., AAA/, AABA, ..., ///9, ///+, ////

Increment by 1 → Increment by AAAB

+ Base-64				a	+	Z	/
				A=0	A=0	A=0	B=1

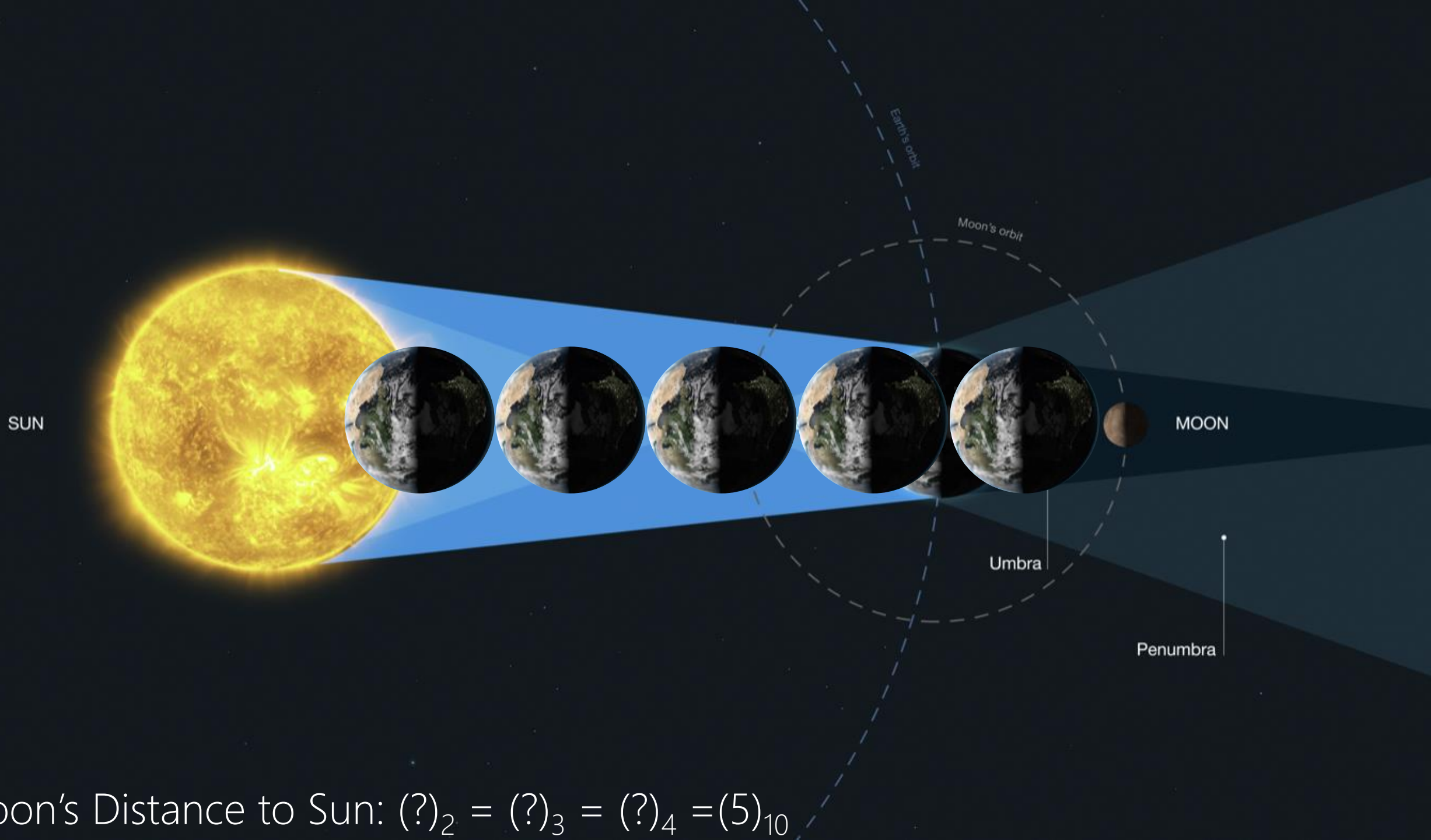
Increment by 1 → Increment by **AAAB**

+ Base-64				a ₌₂₆	+ ₌₆₂	Z ₌₂₅	/ ₌₆₃
				A ₌₀	A ₌₀	A ₌₀	B ₌₁

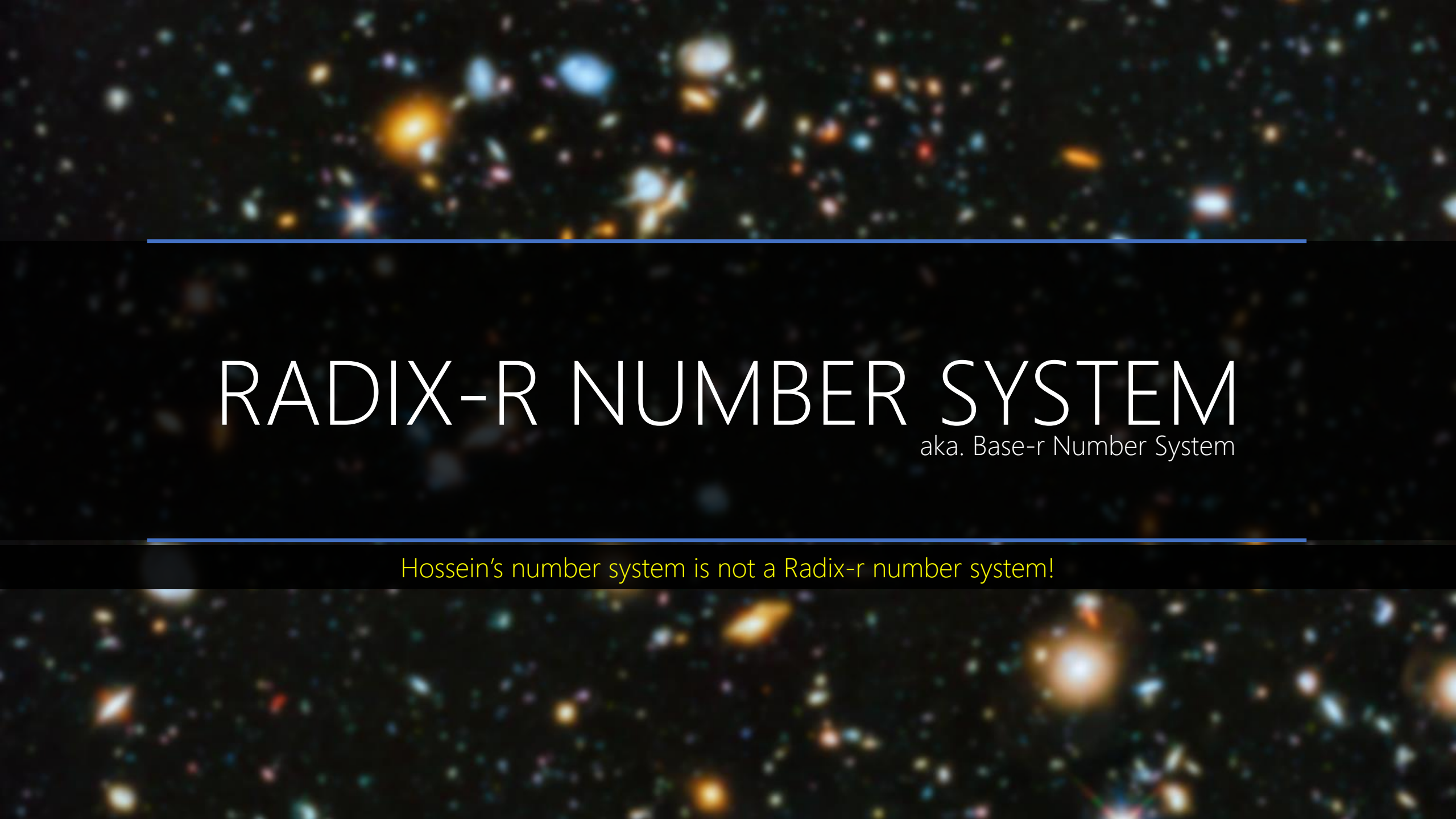
+ Base-64				a ₌₂₆	+ ₌₆₂	B ₌₁	
				A ₌₀	A ₌₀	Z ₌₂₅	/ ₌₆₃
						A ₌₀	B ₌₁
							A ₌₀

$$\frac{64}{64} = 1 \text{ r } 0$$

+ Base-64				a ₌₂₆	+ ₌₆₂	B ₌₁	
				A ₌₀	A ₌₀	Z ₌₂₅	/ ₌₆₃
				a	+	a ₌₂₆	A ₌₀



Moon's Distance to Sun: $(?)_2 = (?)_3 = (?)_4 = (5)_{10}$



RADIX-R NUMBER SYSTEM

aka. Base-r Number System

Hosseini's number system is not a Radix-r number system!

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1} r^{n-1} + d_{n-2} r^{n-2} + \dots + d_i r^i + \dots + d_2 r^2 + d_1 r^1 + d_0 r^0)_{10}$$

where:

r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight (significance) of position i

n = number of digits in N

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1} r^{n-1} + d_{n-2} r^{n-2} + \dots + d_i r^i + \dots + d_2 r^2 + d_1 r^1 + d_0 r^0)_{10}$$

where:

r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight (significance) of position i

n = number of digits in N

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

Min	$= (0_{n-1}0_{n-2} \cdots 0_10_0)_r$	$= (0)_{10}$
Max	$= ((r-1)_{n-1}(r-1)_{n-2} \cdots (r-1)_1(r-1)_0)_r$	$= (r^n - 1)_{10}$
Unit	$= (0_{n-1}0_{n-2} \cdots 0_11_0)_r$	$= (1)_{10}$

where:

- r = radix (base)
- r^i = weight of position i
- n = number of digits in N



FRACTION

The Mystery of Egyptian Fractions (1540 BC)

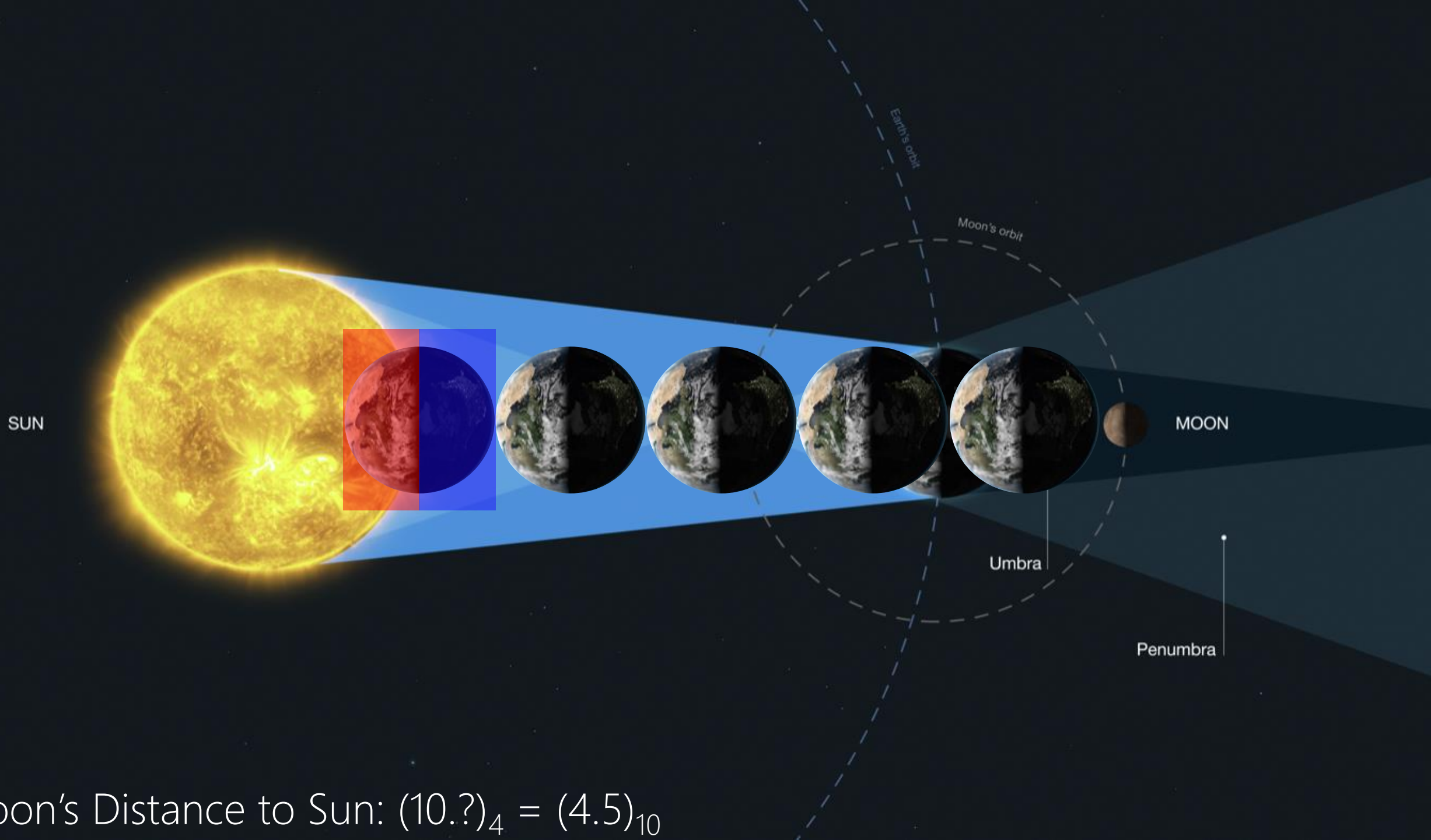
https://en.wikipedia.org/wiki/Egyptian_fraction

https://www.youtube.com/watch?v=_Qr__l-aV6c

$$\frac{1871}{2002} = \frac{1}{2} + \frac{1}{3} + \frac{1}{10} + \frac{1}{812} + \frac{1}{1741740}$$

$$\frac{1871}{2002} = \frac{2}{11} + \frac{7}{13} + \frac{3}{14}$$

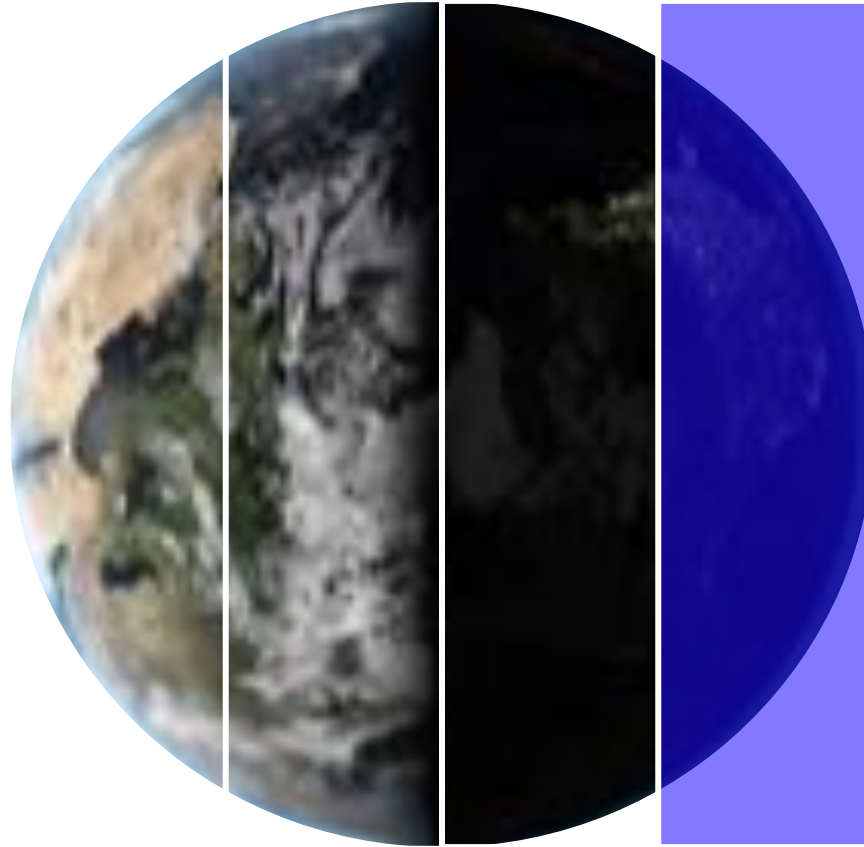




Moon's Distance to Sun: $(10.?)_4 = (4.5)_{10}$

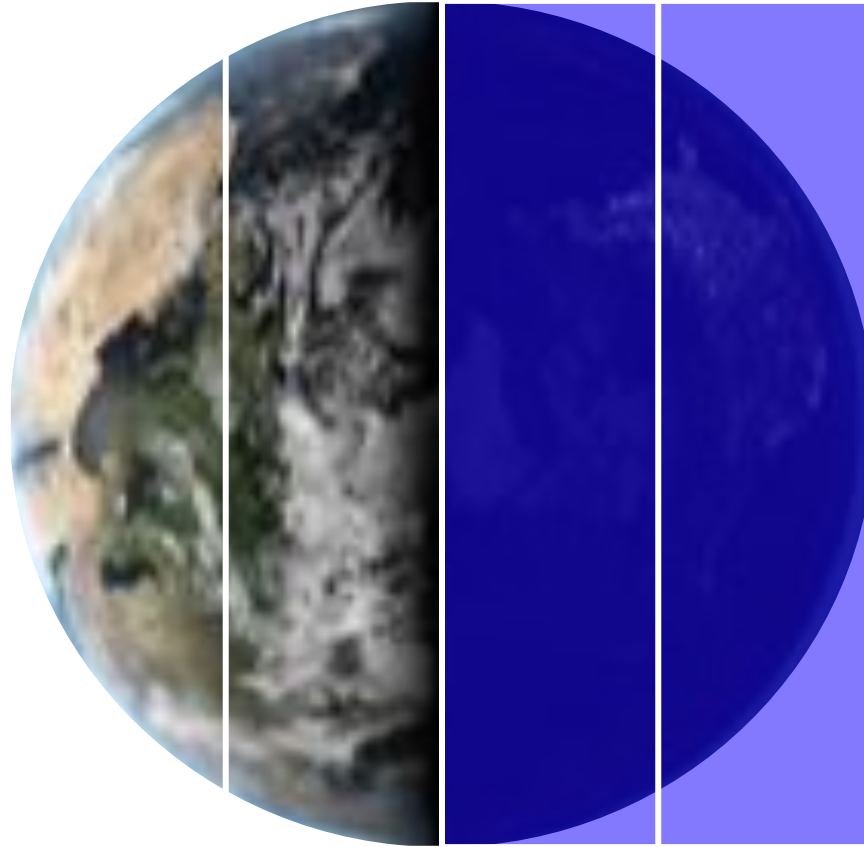


1 Earth

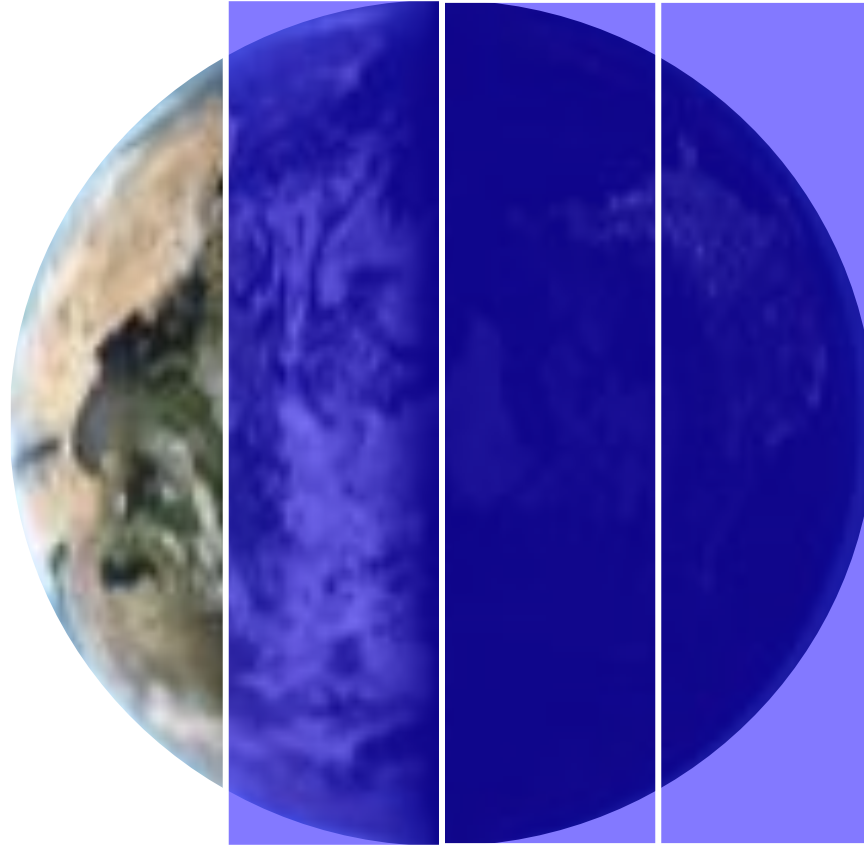


Fraction Point

Radix-4 (Base-4) = $1/4$ Earth = 4^{-1} Earth = $(.1)_4$



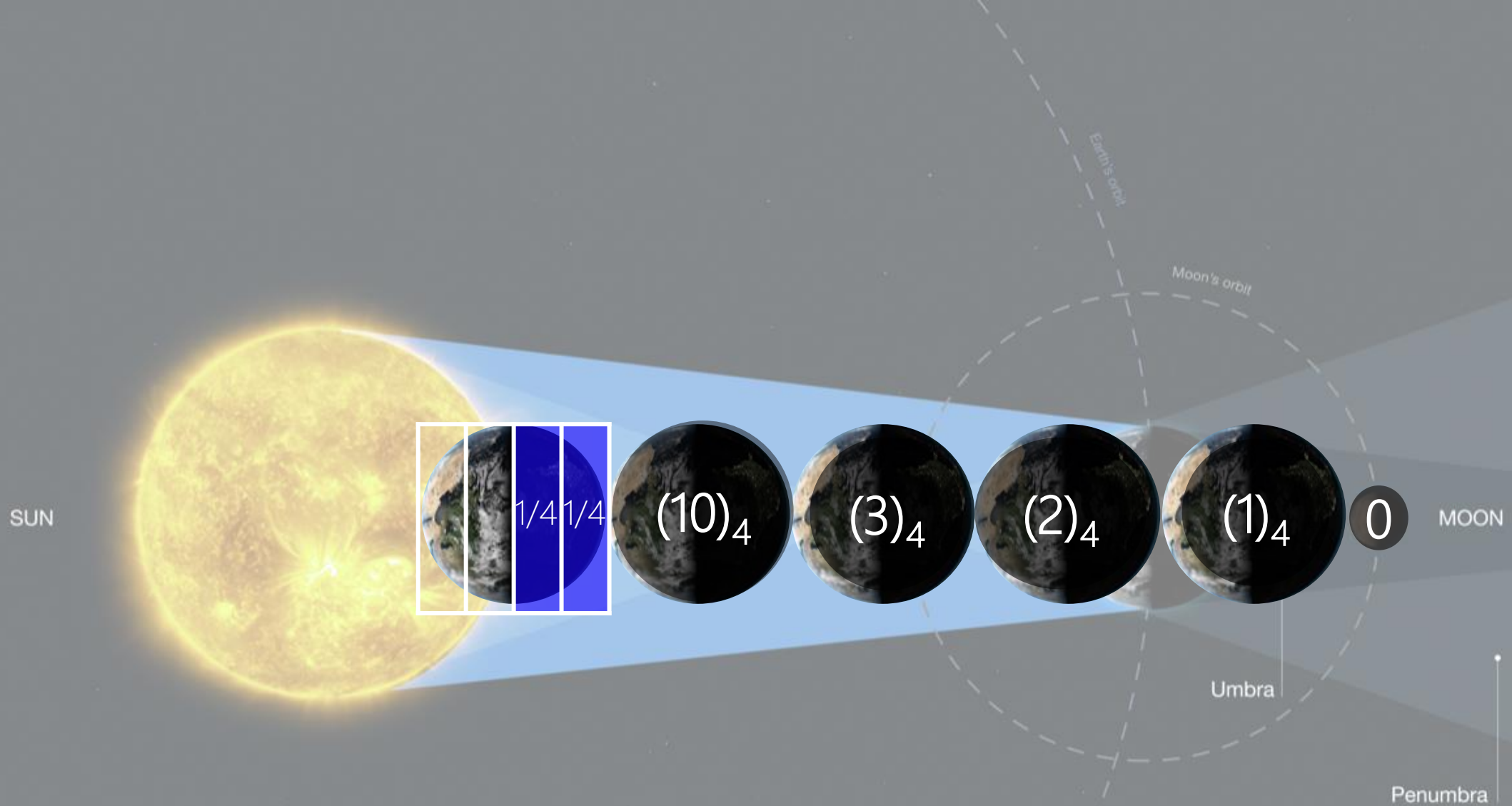
Radix-4 (Base-4) = $2 \times 1/4 \text{ Earth} = 2 \times 4^{-1} \text{ Earth} = (.2)_4$



Radix-4 (Base-4) = $3 \times 1/4 \text{ Earth} = 3 \times 4^{-1} \text{ Earth} = (.3)_4$

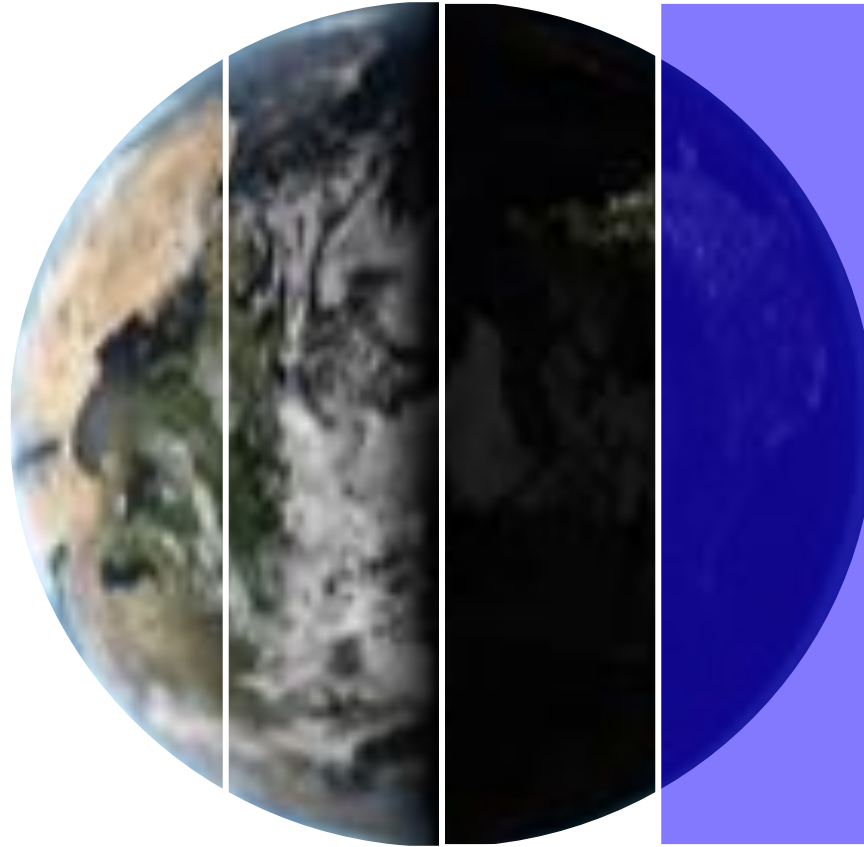


Radix-4 (Base-4) = $4 \times 1/4 \text{ Earth} = 4 \times 4^{-1} \text{ Earth} = (1)_4$

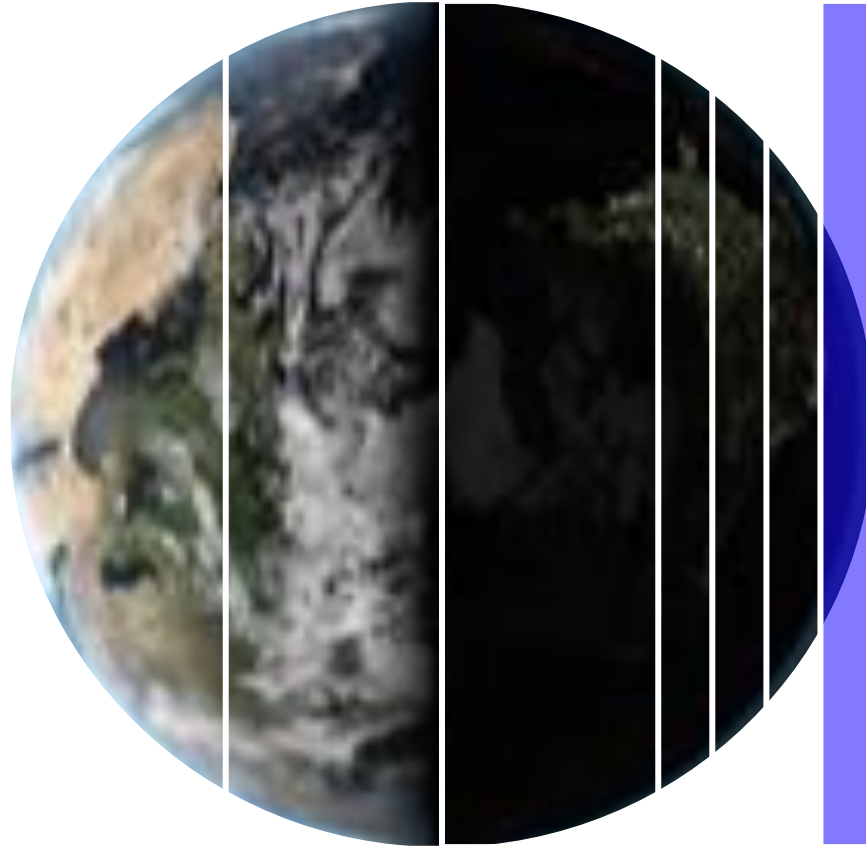


Moon's Distance to Sun in Radix-4: $(10 \cdot 2)_4$

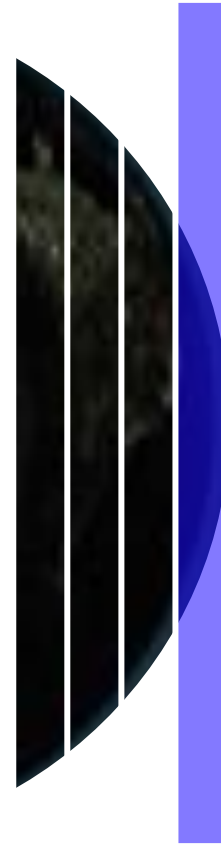
MORE PRECISION



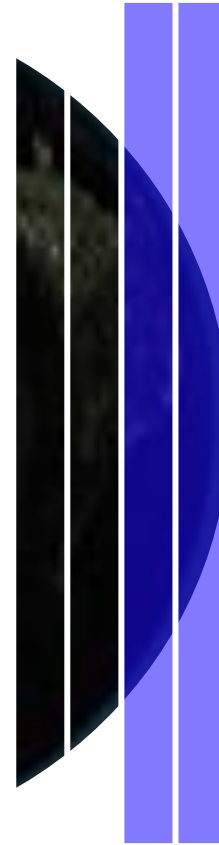
Radix-4 (Base-4) = $1/4$ Earth = 4^{-1} Earth = $(.1)_4$



Radix-4 (Base-4) = $(1/4)/4$ Earth = $1/16$ Earth = 4^{-2} Earth



Radix-4 (Base-4) = $1/16 = 4^{-2}$ Earth = $(.01)_4$



$$\text{Radix-4 (Base-4)} = 2 \times 1/16 = 2 \times 4^{-2} \text{ Earth} = (.02)_4$$

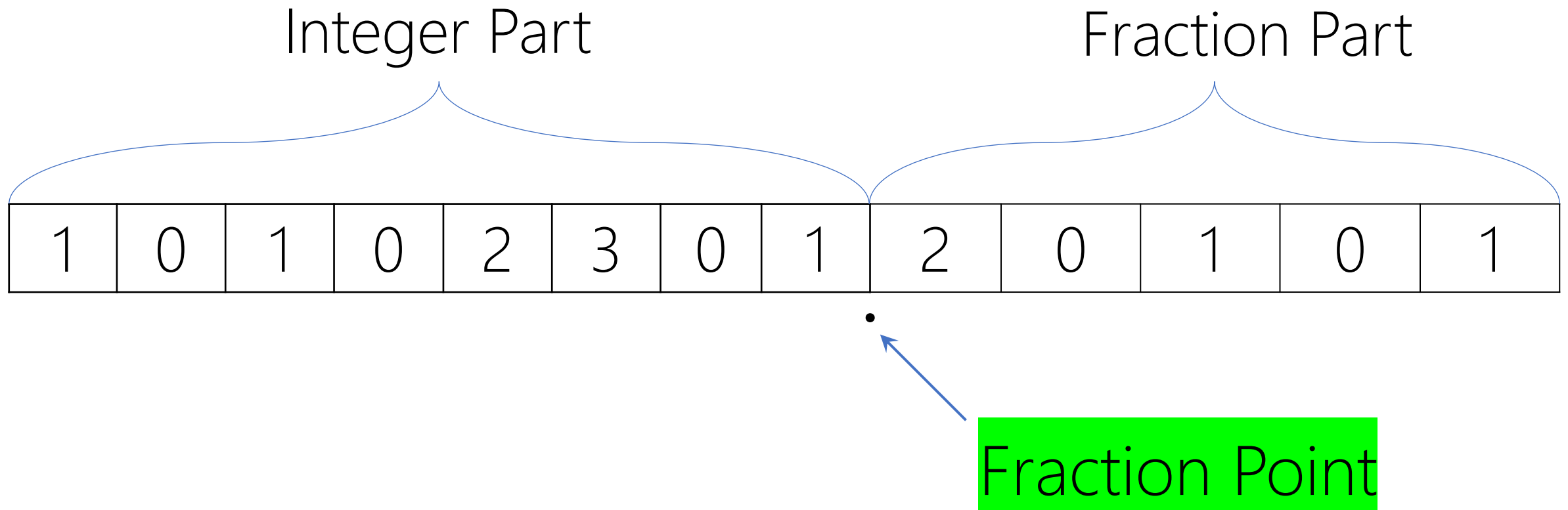


$$\text{Radix-4 (Base-4)} = 3 \times 1/16 = 3 \times 4^{-2} \text{ Earth} = (.03)_4$$



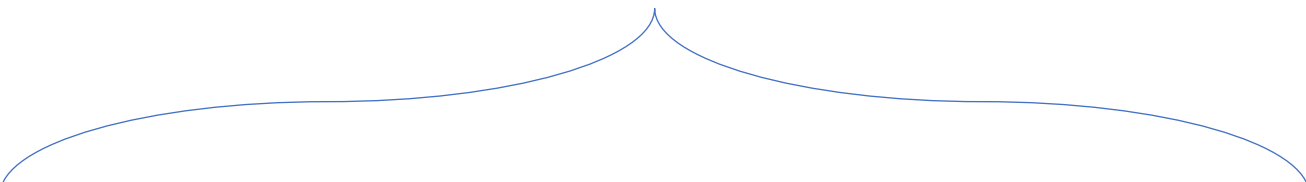
$$\text{Radix-4 (Base-4)} = 4 \times 1/16 = 4 \times 4^{-2} \text{ Earth} = (.1)_4$$





We don't waste a position for the fraction point.
It's location is defined already.

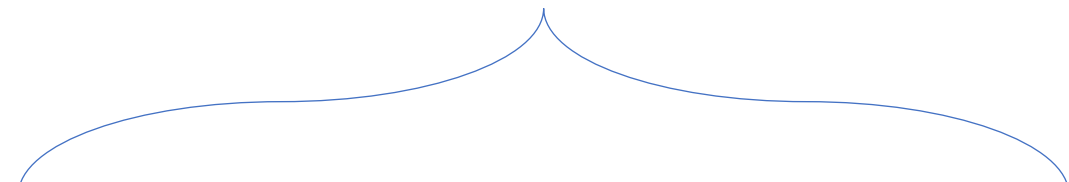
Integer Part



4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0
1	0	1	0	2	3	0	1

.

Fraction Part



4^{-1}	4^{-2}	4^{-3}	4^{-4}	4^{-5}
2	0	1	0	1

Integer Part

4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0
1	0	1	0	2	3	0	1
1 ×16,384	0	1 ×1,024	0	2 ×64	3 ×16	0	1

×
.
Σ

Fraction Part

4^{-1}	4^{-2}	4^{-3}	4^{-4}	4^{-5}
2	0	1	0	1
$\frac{2}{4}$	0	$\frac{1}{64}$	0	$\frac{1}{1,024}$

Integer Part

4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0
1	0	1	0	2	3	0	1
1 ×16,384	0	1 ×1,024	0	2 ×64	3 ×16	0	1
				17,584			

×
.
Σ
.

Fraction Part

4^{-1}	4^{-2}	4^{-3}	4^{-4}	4^{-5}
2	0	1	0	1
$\frac{2}{4}$	0	$\frac{1}{64}$	0	$\frac{1}{1,024}$

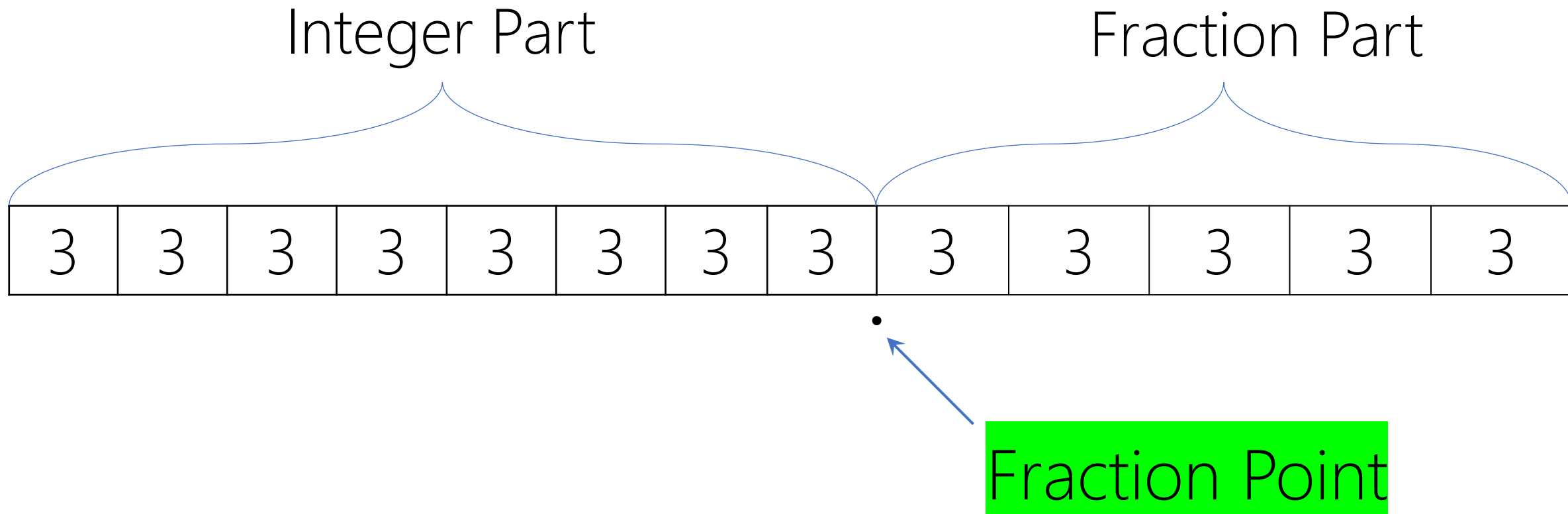
Integer Part

4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0
1	0	1	0	2	3	0	1
1 ×16,384	0	1 ×1,024	0	2 ×64	3 ×16	0	1
				17,584			

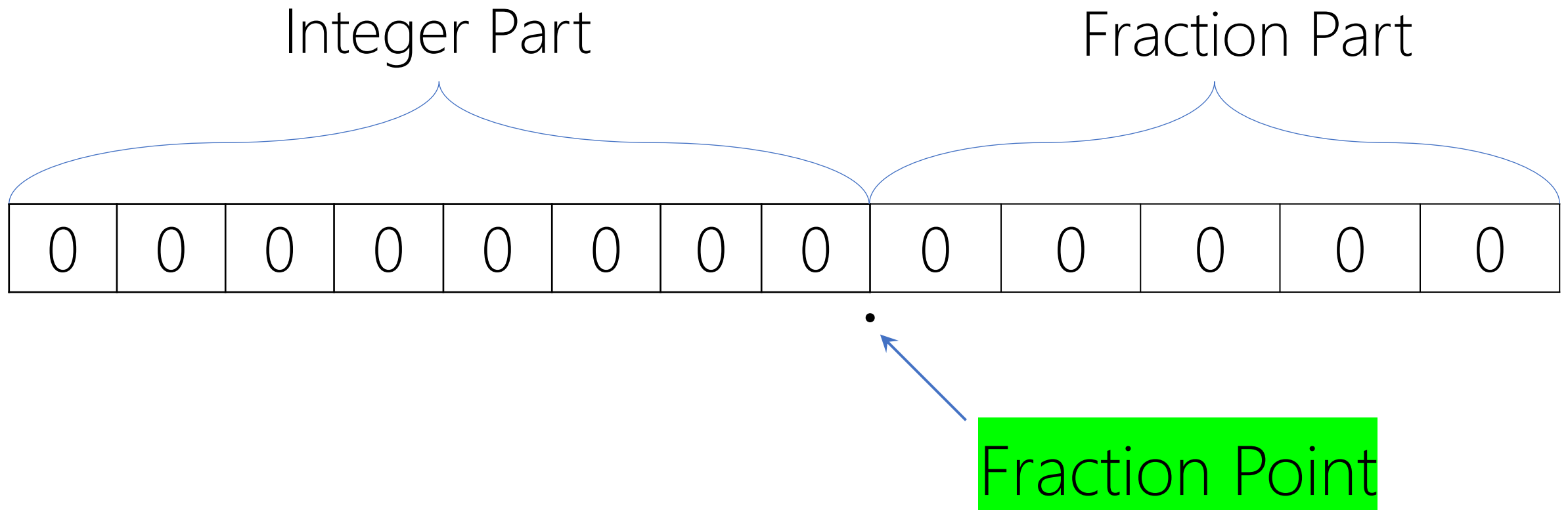
×
.
Σ
.

Fraction Part

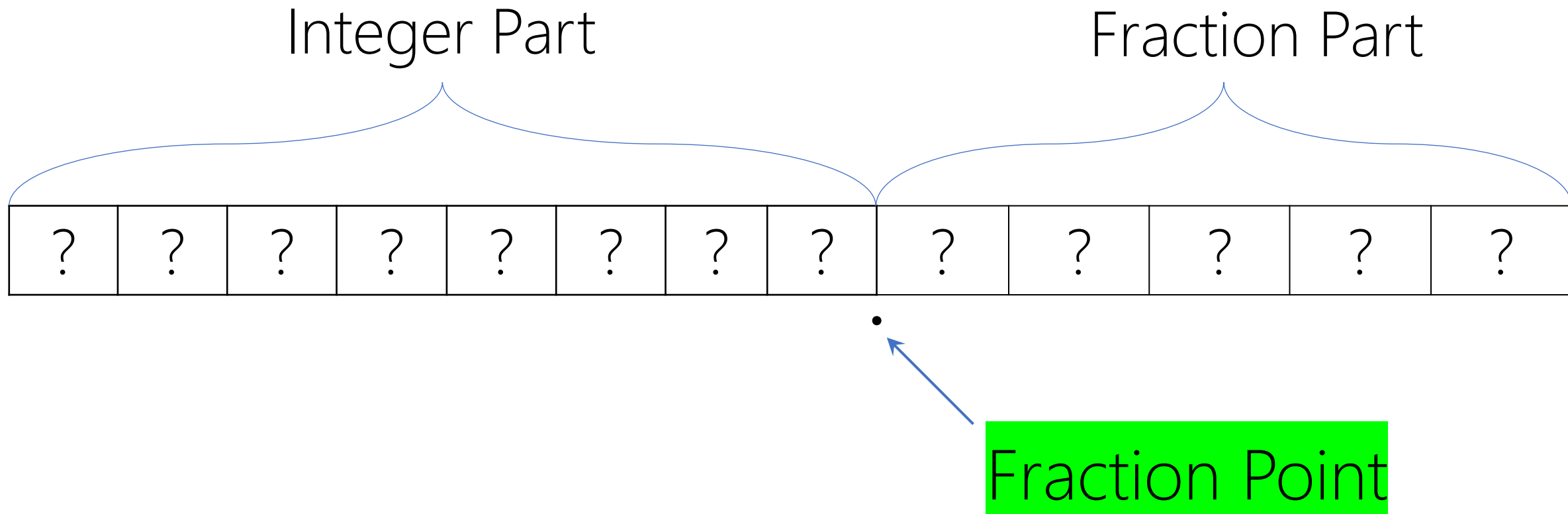
4^{-1}	4^{-2}	4^{-3}	4^{-4}	4^{-5}
2	0	1	0	1
$\frac{2}{4}$	0	$\frac{1}{64}$	0	$\frac{1}{1,024}$
5166015625				



Largest number in base-4 given 8 integer and 5 fraction positions.



Smallest number in base-4 given 8 integer and 5 fraction positions.



Smallest STEP in base-4 given 8 integer and 5 fraction positions?

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1}r^{n-1} + \dots + d_0r^0 \cdot d_{-1}r^{-1} + d_{-2}r^{-2} + \dots + d_{-m}r^{-m})_{10}$$

where:

r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight of position i

n = number of digits in integer part of N

m = number of digits in fraction part of N



Fraction Point

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$\begin{aligned}
 \text{Min} &= (0_{n-1} \cdots 0_1 0_0 \mid 0_{-1} 0_{-2} \cdots 0_{-m-1} 0_{-m})_r = (0.0)_{10} \\
 \text{Max} &= ((r-1)_{n-1} \cdots (r-1)_0 \mid (r-1)_{-1} (r-1)_{-2} \cdots (r-1)_{-m-1} (r-1)_{-m})_r = (r^n - 1 \mid ?)_{10} \\
 \text{Unit} &= (0_{n-1} \cdots 0_1 0_0 \mid 0_{-1} 0_{-2} \cdots 0_{-m-1} 1_{-m})_r = (r^{-m})_{10}
 \end{aligned}$$

where:

r = radix (base)

r^i = weight of position i

n = number of digits in integer part of N

m = number of digits in fraction part of N

Lecture Assignment

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

PRACTICE RADIX-2

Radix-2								
Integer (n=4)					Fraction (m=3)			Radix-10
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1 \cdot 2^{-3} = 1/8 = 0.125$
0	0	0	0	.	0	1	0	$1 \cdot 2^{-2} + 0 \cdot 2^{-3} = 1/4 = 0.25$
0	0	0	0	.	0	1	1	$1 \cdot 2^{-2} + 1 \cdot 2^{-3} = 1/4 + 1/8 = 3/8 = 0.375$
0	0	0	0	.	1	0	0	$1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 0 \cdot 2^{-3} = 1/2 = 0.5$
0	0	0	0	.	1	0	1	$1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} = 1/2 + 1/8 = 5/8 = 0.625$
0	0	0	0	.	1	1	0	$1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 0 \cdot 2^{-3} = 1/2 + 1/4 = 3/4 = 0.75$
0	0	0	0	.	1	1	1	$1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} = 1/2 + 1/4 + 1/8 = 0.875$
0	0	0	1	.	0	0	0	$1 \cdot 2^0 + 0 \cdot 2^{-1} + 0 \cdot 2^{-2} + 0 \cdot 2^{-3} = 1$
0	0	0	1	.	0	0	1	1.125
0	0	0	1	.	0	1	0	1.25
0	0	0	1	.	0	1	1	1.375
0	0	0	1	.	1	0	0	1.5
0	0	0	1	.	1	0	1	1.625
0	0	0	1	.	1	1	0	1.75
0	0	0	1	.	1	1	1	1.875
0	0	1	0	.	0	0	0	2

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

What is the **max** in this system with these spaces?

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

What is the **max** in this system with these spaces?
 $(1111.111)_2 = (15.875)_{10}$

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

A. More precision.

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

A. More precision.

A. More fraction positions.

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

A. More precision.

A. More fraction positions.

A. More in m!

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

- Solution?
- A. More precision.
 - A. More fraction positions.
 - A. More in m! **How much?**

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

B. Find the closest number

Radix-2									
Integer (n=4)					Fraction (m=3)			Radix-10	
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

B. Find the closest number
 $(1.000)_2 = (1)_{10} \Rightarrow \text{Error} = 0.02$
 $(1.001)_2 = (1.125)_{10} \Rightarrow \text{Error} = 0.105$

Radix-2									Radix-10
Integer (n=4)					Fraction (m=3)				
2 ³	2 ²	2 ¹	2 ⁰	.	2 ⁻¹	2 ⁻²	2 ⁻³		
0	0	0	0	.	0	0	0	0	
0	0	0	0	.	0	0	1	1*2 ⁻³ = 1/8 = 0.125	
0	0	0	0	.	0	1	0	1*2 ⁻² + 0*2 ⁻³ = 1/4 = 0.25	
0	0	0	0	.	0	1	1	1*2 ⁻² + 1*2 ⁻³ = 1/4 + 1/8 = 3/8 = 0.375	
0	0	0	0	.	1	0	0	1*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = ½ = 0.5	
0	0	0	0	.	1	0	1	1*2 ⁻¹ + 0*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/8 = 5/8 = 0.625	
0	0	0	0	.	1	1	0	1*2 ⁻¹ + 1*2 ⁻² + 0*2 ⁻³ = 1/2 + 1/4 = 3/4 = 0.75	
0	0	0	0	.	1	1	1	1*2 ⁻¹ + 1*2 ⁻² + 1*2 ⁻³ = 1/2 + 1/4 + 1/8 = 0.875	
0	0	0	1	.	0	0	0	1*2 ⁰ + 0*2 ⁻¹ + 0*2 ⁻² + 0*2 ⁻³ = 1	
0	0	0	1	.	0	0	1	1.125	
0	0	0	1	.	0	1	0	1.25	
0	0	0	1	.	0	1	1	1.375	
0	0	0	1	.	1	0	0	1.5	
0	0	0	1	.	1	0	1	1.625	
0	0	0	1	.	1	1	0	1.75	
0	0	0	1	.	1	1	1	1.875	
0	0	1	0	.	0	0	0	2	

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?
B. Find the closest number
 $(1.000)_2 = (1)_{10} \Rightarrow \text{Error} = 0.02$
 $(1.001)_2 = (1.125)_{10} \Rightarrow \text{Error} = 0.105$

A cosmic background image featuring a dense field of galaxies and stars against a black sky. The galaxies are in various colors, including blue, orange, and white, and are scattered across the frame. Two horizontal blue lines are positioned above and below the central text.

PRACTICE RADIX-4

Radix-4								
Integer (n=4)					Fraction (m=3)			Radix-10
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
...								
3	3	3	3	.	3	3	0	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 0*4 ⁻³ = ?
3	3	3	3	.	3	3	1	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 1*4 ⁻³ = ?
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Radix-4								Radix-10
Integer (n=4)				Fraction (m=3)				
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
...								
3	3	3	3	.	3	3	0	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 0*4 ⁻³ = ?
3	3	3	3	.	3	3	1	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 1*4 ⁻³ = ?
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

Radix-4								Radix-10
Integer (n=4)				Fraction (m=3)				
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
...								
3	3	3	3	.	3	3	0	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 0*4 ⁻³ = ?
3	3	3	3	.	3	3	1	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 1*4 ⁻³ = ?
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Radix-4								Radix-10
Integer (n=4)				Fraction (m=3)				
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
...								
3	3	3	3	.	3	3	0	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 0*4 ⁻³ = ?
3	3	3	3	.	3	3	1	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 1*4 ⁻³ = ?
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

- Solution:
- A. More in m
 - B. Find the closest number

Radix-4								
Integer (n=4)					Fraction (m=3)			Radix-10
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
0	0	0	1	.	0	0	1	1.015625
0	0	0	1	.	0	0	2	1.03125
...								
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

- Solution:
- A. More in m
 - B. Find the closest number
- $(1.001)_4 = (1.015625)_{10} \Rightarrow \text{Error} = 0.004375$
- $(1.002)_4 = (1.03125)_{10} \Rightarrow \text{Error} = 0.01125$

Radix-4								
Integer (n=4)					Fraction (m=3)			Radix-10
4 ³	4 ²	4 ¹	4 ⁰	.	4 ⁻¹	4 ⁻²	4 ⁻³	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	1*4 ⁻³ = 1/64 = 0.015625
0	0	0	0	.	0	0	2	2*4 ⁻³ = 2/64 = 0.03125
0	0	0	0	.	0	0	3	3*4 ⁻³ = 3/64 = 0.046875
0	0	0	0	.	0	1	0	1*4 ⁻² + 0*4 ⁻² = 1/16 = 0.0625
0	0	0	0	.	0	1	1	1*4 ⁻² + 1*4 ⁻² = 1/16 + 1/64 = 0.078125
0	0	0	0	.	0	1	2	1*4 ⁻² + 2*4 ⁻² = 1/16 + 2/64 = 0.09375
0	0	0	0	.	0	1	3	1*4 ⁻² + 3*4 ⁻² = 1/16 + 3/64 = 0.109375
0	0	0	0	.	0	2	0	2*4 ⁻² + 0*4 ⁻² = 2/16 = 0.125
...								
0	0	0	0	.	3	3	3	3*4 ⁻¹ + 3*4 ⁻² + 3*4 ⁻³ = 0.984375
0	0	0	1	.	0	0	0	1
0	0	0	1	.	0	0	1	1.015625
0	0	0	1	.	0	0	2	1.03125
...								
3	3	3	3	.	3	3	2	3*4 ³ + 3*4 ² + 3*4 ¹ +3*4 ⁰ +3*4 ⁻¹ + 3*4 ⁻² + 2*4 ⁻³ = ?
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Solution:

A. More in m

B. Find the closest number

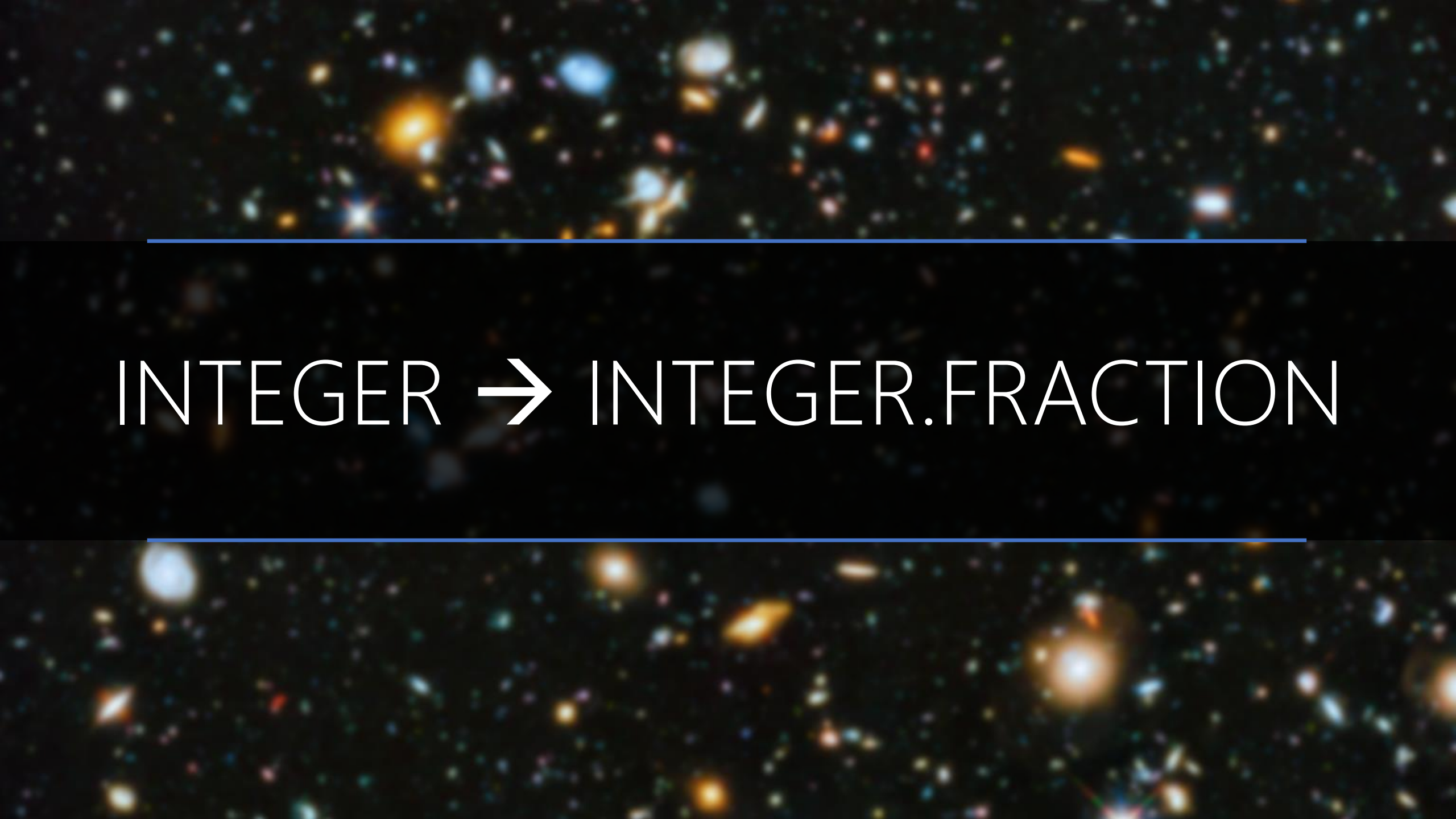
$(1.001)_4 = (1.015625)_{10} \Rightarrow \text{Error} = 0.004375$

$(1.002)_4 = (1.03125)_{10} \Rightarrow \text{Error} = 0.01125$

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines are positioned above and below the central text.

PRACTICE RADIX-[8,10,16]

At Home

A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, red) against a black space. Two horizontal blue lines are positioned above and below the central text.

INTEGER → INTEGER.FRACTION

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

INTEGER.FRACTION → POSITIVE

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

INTEGER.FRACTION → NEGATIVE?



BE PATIENT ...

A deep space image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

CONVERSION

From Base-r to Base-r'

Diagram illustrating a transformation from a 5x5 grid of white squares to a 5x5 grid of blue squares, with a red arrow pointing from the first grid to the second. The grids are labeled with subscripts r and r' respectively.